

Security and Protection of Systems

Lecture 08 *Symmetric Cryptography*

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Outlines

- Advanced Encryption Standard
- Double and Triple DES
- Modes of Operation

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- Modes of Operation

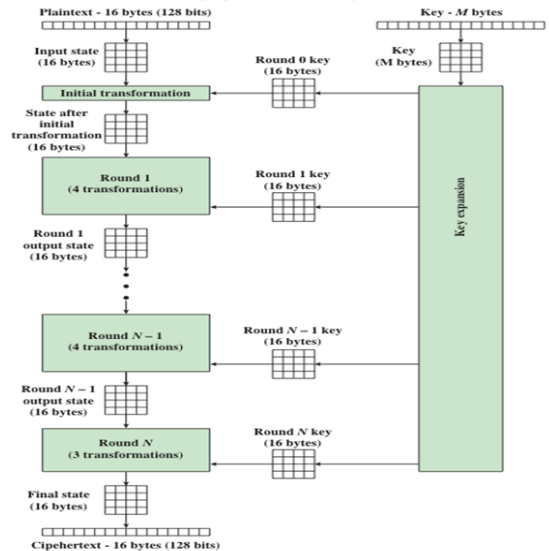
Advanced Encryption Standard (AES)

- The **AES**, also known by its original name **Rijndael** is a specification for the **encryption** of electronic data established by the U.S. (NIST) in 2001.
- Rijndael is a family of ciphers with different key and block sizes. For AES, NIST selected three members of the Rijndael family, each with a block size of 128 bits, but three different key lengths: 128, 192 and 256 bits.
- The algorithm described by AES is a **symmetric-key algorithm**, meaning the same key is used for both encrypting and decrypting the data.

Key Size (words/bytes/bits)	4/16/128	6/24/192	8/32/256
Plaintext Block Size (words/bytes/bits)	4/16/128	4/16/128	4/16/128
Number of Rounds	10	12	14
Round Key Size (words/bytes/bits)	4/16/128	4/16/128	4/16/128
Expanded Key Size (words/bytes)	44/176	52/208	60/240

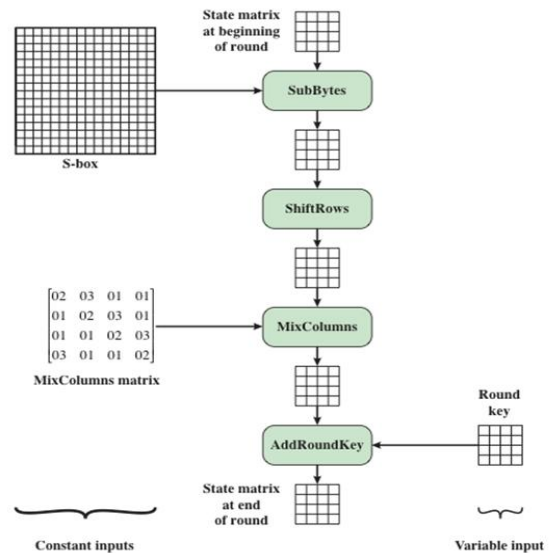
The overall structure of the AES encryption process.

- First $N - 1$ rounds consist of four transformation functions: [SubBytes](#), [ShiftRows](#), [MixColumns](#), and [AddRoundKey](#).
- Final round contains only three transformations, and a initial single transformation ([AddRoundKey](#)) before the first round
- Each transformation takes 4×4 matrices as input and produces a 4×4 matrix as output.
- The key expansion function generates $N+1$ round keys, each of which is a distinct 4×4 matrix.



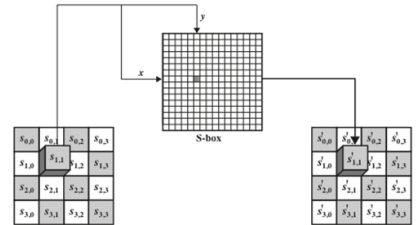
Single Round of AES

- SubBytes** - a non-linear substitution step where each byte is replaced with another according to a lookup table.
- ShiftRows** - a transposition step where the last three rows of the state are shifted cyclically a certain number of steps.
- MixColumns** - a linear mixing operation which operates on the columns of the state, combining the four bytes in each column.
- AddRoundKey** - each byte of the state is combined with a byte of the round key using bitwise xor.



AES Substitution

- Each byte in the *state* array is replaced with a SubByte using an 8-bit **substitution box**.
- This operation provides the non-linearity in the **cipher**.
- The S-box used is derived from the **multiplicative inverse** over finite field known to have good non-linearity properties.
- To avoid attacks based on simple algebraic properties, the S-box is constructed by combining the inverse function with an invertible **affine transformation**.
- The S-box is also chosen to avoid any fixed points and also any opposite fixed points.
- While performing the decryption, the **InvSubBytes** step (**the inverse of SubBytes**) is used, which requires first taking the inverse of the **affine transformation** and then finding the multiplicative inverse.



AES S-box

		y															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
x	0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
	1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
	2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
	3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
	4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
	5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
	6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
	7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
	8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
	9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
	A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
	B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
	C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
	D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
	E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
	F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

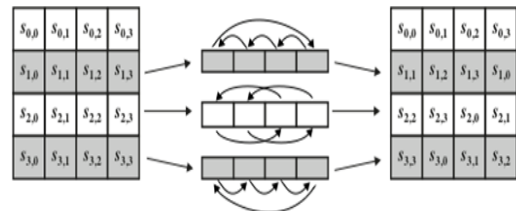
For example, the hexadecimal value {95} references row 9, column 5 of the S-box, which contains the value {2A}. Accordingly, the value {95} is mapped into the value {2A}.

AES Inverse S-box

		y															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
x	0	52	09	6A	D5	30	36	A5	38	BF	40	A3	9E	81	F3	D7	FB
	1	7C	E3	39	82	9B	2F	FF	87	34	8E	43	44	C4	DE	E9	CB
	2	54	7B	94	32	A6	C2	23	3D	EE	4C	95	0B	42	FA	C3	4E
	3	08	2E	A1	66	28	D9	24	B2	76	5B	A2	49	6D	8B	D1	25
	4	72	F8	F6	64	86	68	98	16	D4	A4	5C	CC	5D	65	B6	92
	5	6C	70	48	50	FD	ED	B9	DA	5E	15	46	57	A7	8D	9D	84
	6	90	D8	AB	00	8C	BC	D3	0A	F7	E4	58	05	B8	B3	45	06
	7	D0	2C	1E	8F	CA	3F	0F	02	C1	AF	BD	03	01	13	8A	6B
	8	3A	91	11	41	4F	67	DC	EA	97	F2	CF	CE	F0	B4	E6	73
	9	96	AC	74	22	E7	AD	35	85	E2	F9	37	E8	1C	75	DF	6E
	A	47	F1	1A	71	1D	29	C5	89	6F	B7	62	0E	AA	18	BE	1B
	B	FC	56	3E	4B	C6	D2	79	20	9A	DB	C0	FE	78	CD	5A	F4
	C	1F	DD	A8	33	88	07	C7	31	B1	12	10	59	27	80	EC	5F
	D	60	51	7F	A9	19	B5	4A	0D	2D	E5	7A	9F	93	C9	9C	EF
	E	A0	E0	3B	4D	AE	2A	F5	B0	C8	EB	BB	3C	83	53	99	61
	F	17	2B	04	7E	BA	77	D6	26	E1	69	14	63	55	21	0C	7D

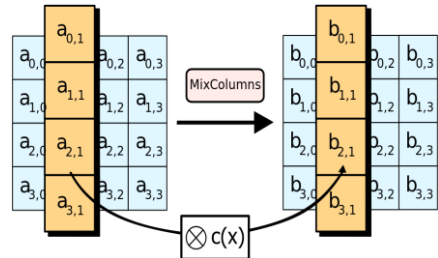
Shift Row Rationale

- Operates on the rows of the state; it cyclically shifts the bytes in each row by a certain **offset**.
- The first row is left unchanged.
- Each byte of the second row is shifted one to the left. Similarly, the third and fourth rows are shifted by offsets of two and three respectively.
- In this way, each column of the output state of the **ShiftRows** step is composed of bytes from each column of the input state.
- The importance of this step is to avoid the columns being encrypted independently, in which case AES would degenerate into four independent block ciphers.



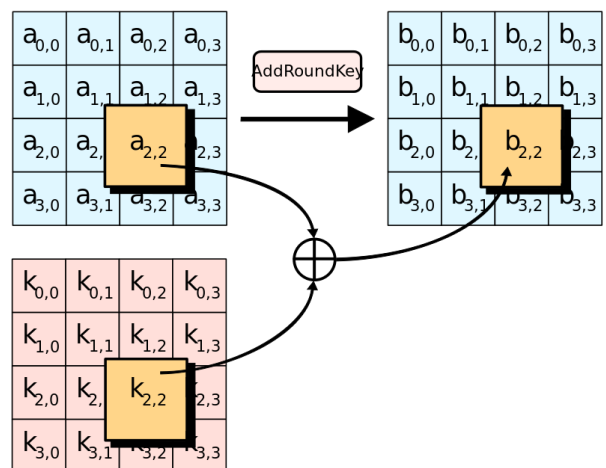
Mix Columns Rationale

- In the MixColumns step, the four bytes of each column of the state are combined using an invertible **linear transformation**.
- The MixColumns function takes four bytes as input and outputs four bytes, where each input byte affects all four output bytes.
- Together with ShiftRows, MixColumns provides diffusion in cipher
- During this operation, each column is transformed using a fixed matrix (matrix left-multiplied by column gives new value of column in the state)



AddRoundKey Transformation

- In the AddRoundKey step, the subkey is combined with the state. For each round, a subkey is derived from the main **key** using **Rijndael's key schedule**; each subkey is the same size as the state. The subkey is added by combining each byte of the state with the corresponding byte of the subkey using bitwise **XOR**.



AES Security

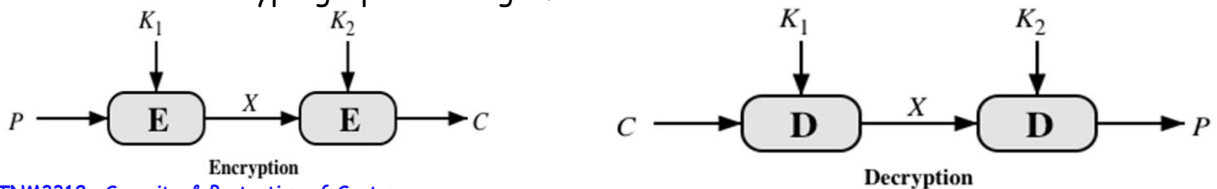
1. The design and strength of all key lengths of the AES algorithm (i.e., 128, 192 and 256) are sufficient to protect classified information up to the SECRET level.
2. TOP SECRET information will require use of either the 192 or 256 key lengths. The implementation of AES in products intended to protect national security systems and/or information must be reviewed and certified by NSA prior to their acquisition and use
3. AES has 10 rounds for 128-bit keys, 12 rounds for 192-bit keys, and 14 rounds for 256-bit keys.

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Double DES

- Because of its vulnerability to brute-force attack, symmetric cipher, has been largely replaced by stronger encryption schemes.
- Use multiple encryption with DES and multiple keys.
- The simplest form of multiple encryption has two encryption stages and two keys
- Given a plaintext P and two encryption keys K_1 and K_2 , ciphertext C is generated as $C = E(K_2, E(K_1, P))$, $P = D(K_1, D(K_2, C))$
- Double DES involves a key length of $56 * 2 = 112$ bits, resulting in a dramatic increase in cryptographic strength.

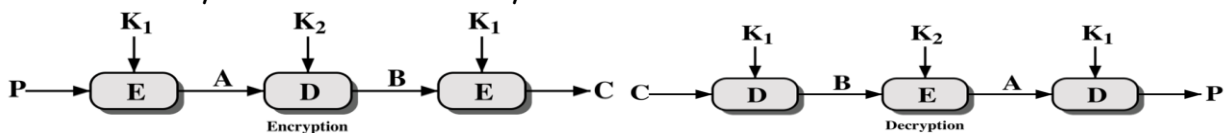


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Triple-DES with Two-Keys

- Three stages of encryption with three different keys.
- This raises the cost of the meet-in-the-middle attack to 2^{112} , which is beyond what is practical now and far into the future.
 - $C = E(K_1, D(K_2, E(K_1, P)))$
 - $P = D(K_1, E(K_2, D(K_1, C)))$
- However, it has the drawback of requiring a key length of $56 * 3 = 168$ bits, which may be somewhat unwieldy.



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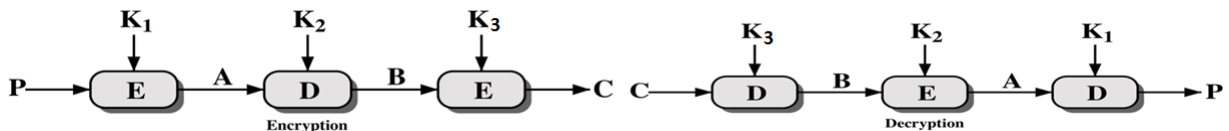
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Triple DES with Three Keys

- Although the attacks just described appear impractical, anyone using two-key 3DES may feel some concern. Thus, many researchers now feel that three-key 3DES is the preferred alternative (e.g., [KALI96a]). Three-key 3DES has an effective key length of 168 bits and is defined as

- $C = E(K_3, D(K_2, E(K_1, P)))$

- $P = D(K_3, E(K_2, D(K_1, P)))$



Outlines

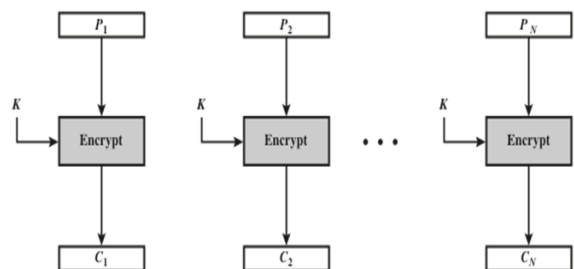
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Modes of Operation

- A block cipher takes a fixed-length block **b-bit** and a key as input and produces a **b-bit** ciphertext block.
- If the amount of plaintext to be encrypted is greater than **b-bit**, then the block cipher can still be used by breaking the plaintext up into **b-bit** blocks.
- When **n** blocks are encrypted using the same key, a number of **security issues arise**.
- Mode of operation is a technique for enhancing the effect or adapting of a cryptographic algorithm, such as applying a block cipher to a sequence of data blocks or a **data stream**.
- Modes are intended to cover a wide of applications for which a block cipher could be used.
- These modes are intended for use with any symmetric block cipher (triple DES and AES).

1- Electronic Codebook (ECB) mode

- Plaintext is handled one block at a time and each block is encrypted using the same key.
- The term codebook is used because, for a given key, there is a unique ciphertext for every **b-bit** block of plaintext.

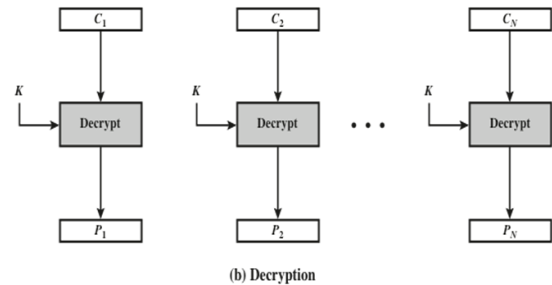


(a) Encryption

- Therefore, we can imagine a gigantic codebook in which there is an entry for every possible **b-bit** plaintext pattern showing its corresponding ciphertext.
- For a message longer than **b-bit**, the procedure is simply to break the message into **b-bit** blocks, padding the last block if necessary.

1- Electronic Codebook (ECB) mode

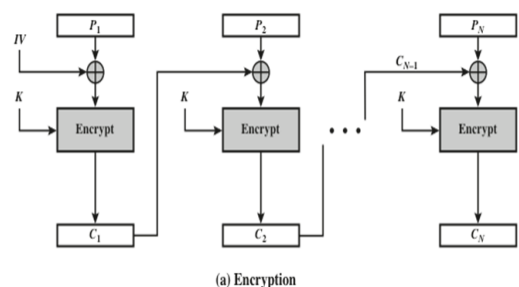
- Decryption is performed one block at a time, always using the same key.
- ECB method is ideal for a short amount of data, such as an encryption key. Thus, if you want to transmit a DES or AES key securely, ECB is the appropriate mode to use.



- The most significant characteristic of ECB is that if the same b -bit block of plaintext appears more than once in the message, it always produces the same ciphertext.
- For lengthy messages, the ECB mode may not be secure. If the message is highly structured, it may be possible for a cryptanalyst to exploit these regularities.

2- Cipher block chaining (CBC) mode

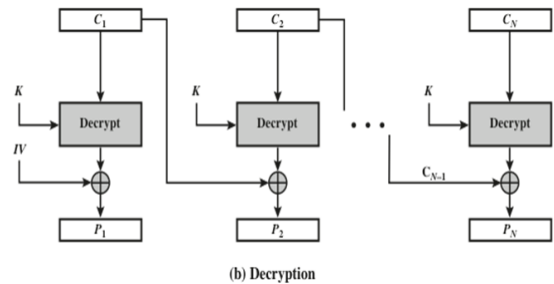
- CBC overcomes the security deficiencies of ECB, if the same plaintext block repeated, produces different ciphertext blocks.
- In CBC, the input to the encryption algorithm is the XOR of the current plaintext block and the preceding ciphertext block; the same key is used for each block.



- In effect, we have chained together the processing of the sequence of plaintext blocks.
- The input to the encryption function for each plaintext block bears no fixed relationship to the plaintext block. Therefore, repeating patterns of b bits are not exposed. As with the ECB mode, the CBC mode requires that the last block be padded to a full b bits if it is a partial block.

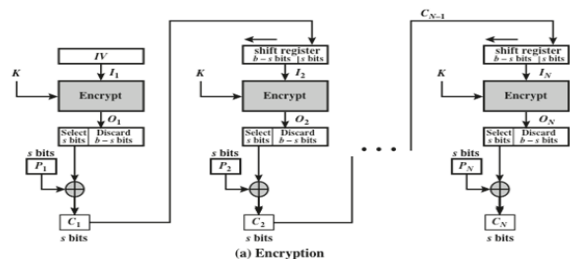
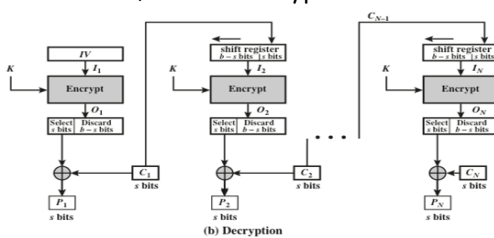
2- Cipher block chaining (CBC) mode

- On decryption, the IV is XORed with the output of the decryption algorithm to recover the first block of plaintext.
- IV is a data block that is the same size as the cipher block, the IV must be known to both the sender and receiver but be unpredictable by a third party.
- In particular, for any given plaintext, it must not be possible to predict the IV that will be associated to the plaintext in advance of the generation of the IV.
- For maximum security, the IV should be protected against unauthorized changes.
- To produce the first block of ciphertext, an initialization vector (IV) is XORed with the first block of plaintext.



3- Cipher feedback (CFB) Mode

- The input to the encryption function is a b -bit shift register that is initially set to some IV.
- The leftmost s bits of the output of the encryption function are XORed with the first segment of plaintext P_1 to produce the first unit of ciphertext C_1 , which is then transmitted.
- The contents of the shift register are shifted left by s bits, and C_1 is placed in the rightmost s bits of the shift register.
- For **decryption**, the same scheme is used, except that the received ciphertext unit is XORed with the output of the encryption function to produce the plaintext unit. Note that it is the encryption function that is used, not the decryption function.

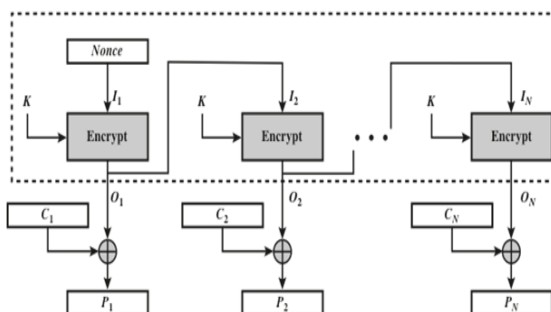


3- Cipher feedback (CFB) Mode

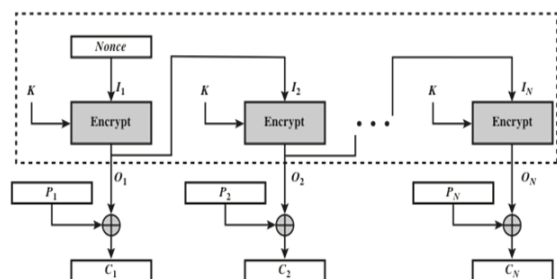
- CFB can be viewed as a stream cipher, it does not conform to the typical construction of a stream cipher.
- In a typical stream cipher, the cipher takes as input some initial value and a key and generates a stream of bits, which is then XORed with the plaintext bits. In the case of CFB, the stream of bits that is XORed with the plaintext also depends on the plaintext.
- In CFB encryption, like CBC encryption, the input block to each forward Cipher function (except the first) depends on the result of the previous forward Cipher function; therefore, multiple forward cipher operations cannot be performed in parallel.
- In CFB decryption, the required forward cipher operations can be performed in parallel if the input blocks are first constructed (in series) from the IV and the ciphertext.

4- Output Feedback (OFB) mode

- For OFB, the output of encryption is fed back to become input for encrypting the next block.
- In CFB, the output of XOR unit is fed back to become input for encrypting the next block.
- The other difference is that the OFB mode operates on full blocks of plaintext and ciphertext, whereas CFB operates on an s -bit subset.



(b) Decryption



(a) Encryption

4- Output Feedback (OFB) mode

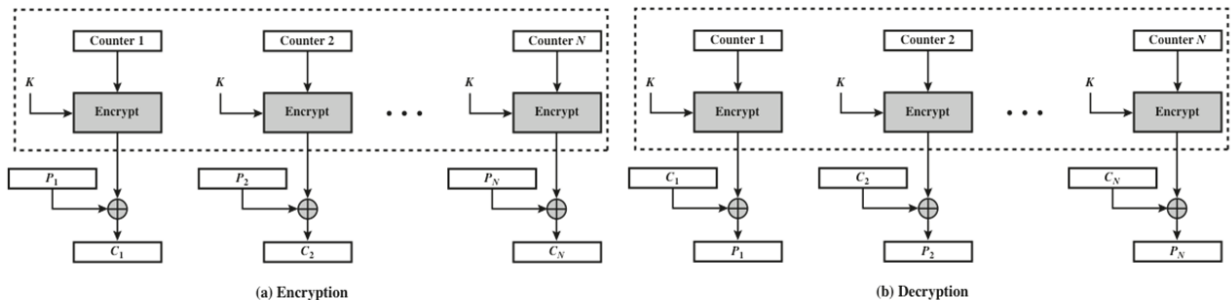
- IV must be a nonce; that is, the IV must be unique to each execution of the encryption operation. The reason for this is that the sequence of encryption output blocks, O_i , depends only on the key and the IV and does not depend on the plaintext.
- Therefore, for a given key and IV, the stream of output bits used to XOR with the stream of plaintext bits is fixed. If two different messages had an identical block of plaintext in the identical position, then an attacker would be able to determine that portion of the O_i stream.
- One advantage of the OFB method is that bit errors in transmission do not propagate.
- The disadvantage of OFB is that it is more vulnerable to a message stream modification attack than is CFB.

counter (CTR) mode

- Although interest in the CTR mode has increased recently with applications to ATM network security and IP sec.
- A counter equal to the plaintext block size is used.
- The only requirement is that the counter value must be different for each plaintext block that is encrypted.
- Typically, the counter is initialized to some value and then incremented by 1 for each subsequent block (modulo 2^b , where b is the block size).

counter (CTR) mode

- For encryption, the counter is encrypted and then XORed with the plaintext block to produce the ciphertext block; there is no chaining.
- For decryption, the same sequence of counter values is used, with each encrypted counter XORed with a ciphertext block to recover the corresponding plaintext block.
- Thus, the initial counter value must be made available for decryption.



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Mode	Description	Typical Application
Electronic Codebook (ECB)	Each block of plaintext bits is encoded independently using the same key.	•Secure transmission of single values (e.g., an encryption key)
Cipher Block Chaining (CBC)	The input to the encryption algorithm is the XOR of the next block of plaintext and the preceding block of ciphertext.	•General-purpose block-oriented transmission •Authentication
Cipher Feedback (CFB)	Input is processed s bits at a time. Preceding ciphertext is used as input to the encryption algorithm to produce pseudorandom output, which is XORed with plaintext to produce next unit of ciphertext.	•General-purpose stream-oriented transmission •Authentication
Output Feedback (OFB)	Similar to CFB, except that the input to the encryption algorithm is the preceding encryption output, and full blocks are used.	•Stream-oriented transmission over noisy channel (e.g., satellite communication)
Counter (CTR)	Each block of plaintext is XORed with an encrypted counter. The counter is incremented for each subsequent block.	•General-purpose block-oriented transmission •Useful for high-speed requirements

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Thanks